MIS-6900

DEEP LEARNING AND TEXT ANALYTICS

AQUA-NET: A DEEP DIVE INTO AQUATIC LIFE

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TABLE OF CONTENT

INTRODUCTION

1

Objective:

Classify aquatic life using CNNs to explore their applications in varied industries.

2

Techniques:

Use data augmentation (rotation, flipping, scaling) and transfer learning to enhance model accuracy and generalization.

3

Project Goal:

Achieve high classification performance and provide a practical case study for optimizing CNNs for real-world image datasets.

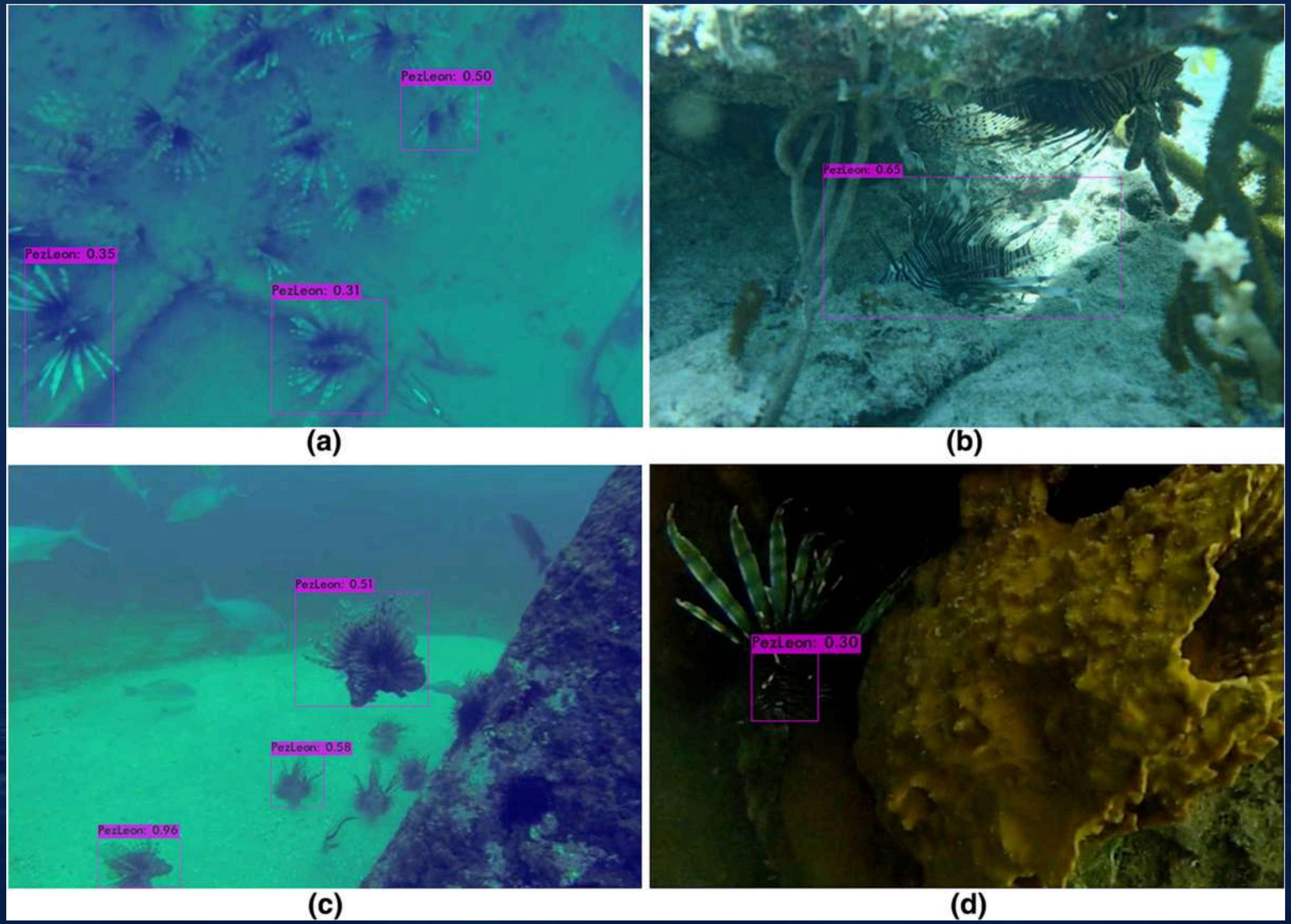
INTRODUCTION

Using AquaNet for Automated Aquatic Species identification with CNNs:

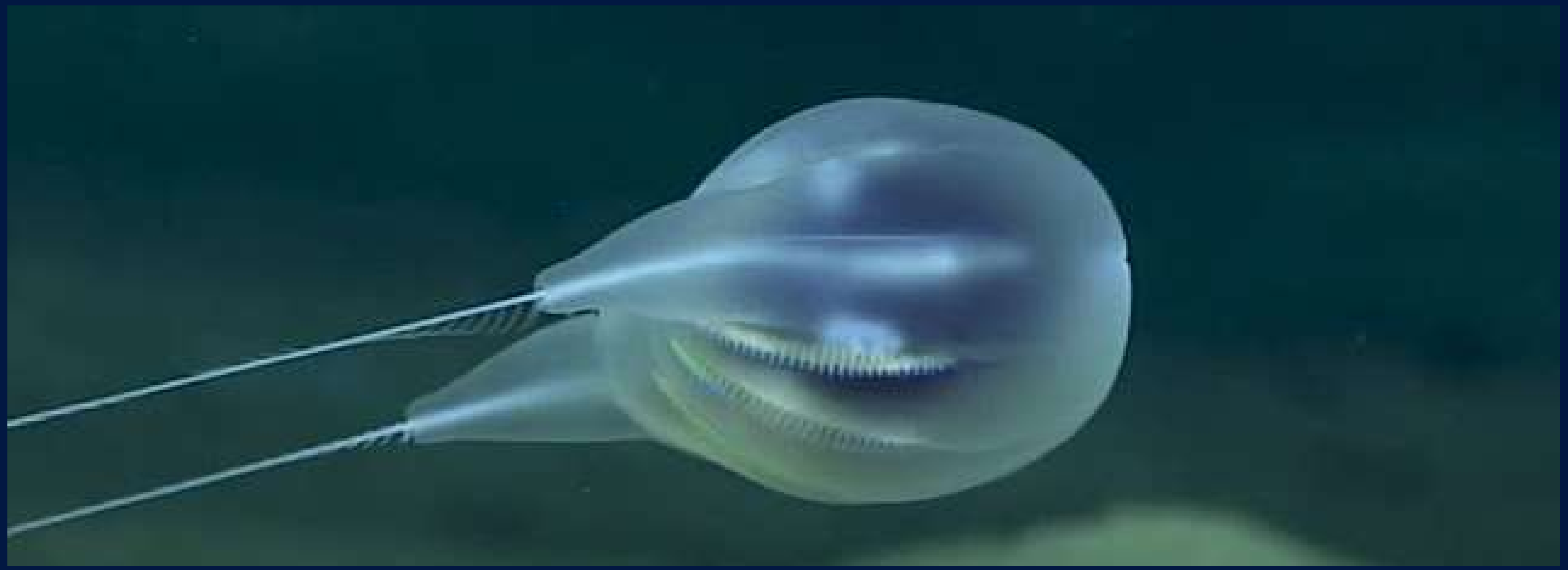
- Fulfils the challenge of vastness and diversity of aquatic ecosystems.
- Benefits utilization of complexities and demands to build efficient tools and methods.
- Plays a crucial role in multiple areas including research, conservation, and education.
- Advancements in deep learning and CNNs revolutionize image recognition, enabling higher accuracy in small sets of species identification.

THE NEED FOR EFFICIENT AQUATIC LIFE IDENTIFICATION

Traditional marine species identification



Real-Time Image Detection and Classification



New species of Duobrachium sparksae – a strange, gelatinous species of ctenophore (solely on high-definition video footage captured at the bottom of the ocean)

Rare and bizarre tentacle-trailing sea creature - sea pen in the Pacific Ocean. (Image credit: Ocean Exploration Trust, NOAA & Oregon State University/Thurber)

PROBLEM STATEMENT

Classifying Aquatic Wildlife

- Traditional methods of species identification can be time-consuming, labor-intensive, and requires expert knowledge.
- There is a growing need for automated and efficient methods to support large-scale monitoring and conservation efforts.

DATASET & DATA PREPARATION

- Source: Kaggle dataset of sea animals featuring images of 23 classes of aquatic life.

- Preparation Steps:
  - Cleaning: Eliminating all irrelevant or corrupted images.
  - Resizing: All images adjusted to a uniform size of 255px.
  - Normalization: Pixel values standardized to a common range (0-1) to enhance model training.

PROJECT FOCUS

3/23 Classes

- Chosen classes include identification of Dolphins, jellyfish, and sea turtles.
- They act as the key indicators of ecological Importance of ocean health.
- Offers visual diversity of varied shapes, textures and features for effective model training.
- These categories offer practical relevance with common encounters in conservation and research in multiple geographical areas.

EXPLORATORY DATA ANALYSIS

Understanding the Data: Insights and Visualizations

- Data Inspection:
  - Analyzed visual traits like resolutions, lighting, aspect ratios, and complex backgrounds.
- Augmentation & Preprocessing:
  - Balanced classes and preserved key features, ensuring accurate feature extraction.
- Model Optimization:
  - Achieved 94.92% accuracy with convolution layers, batch normalization, augmentation and transfer learning.
- Evaluation:
  - Used a confusion matrix and validation loss analysis to assess classification performance.

Model Architecture: Convolutional Neural Network (CNN)

The Power of CNNs for Image Classification

- CNN Architecture: (3500 image total)
    - Input layer: 128x128
    - Hidden layers: Utilize ReLU and Maxpooling.
- Output layer:
    - Softmax function for multi-class classification.
- Training:
    - Data split: 70% (2470) for training, (531) 15% test and (529) 15% for validation
    - Optimizer: Adam
    - Loss function: Cross-entropy.

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TRAIN SET:
  Turtle_Tortoise: 1332 images
  Dolphin: 547 images
  Jelly Fish: 591 images

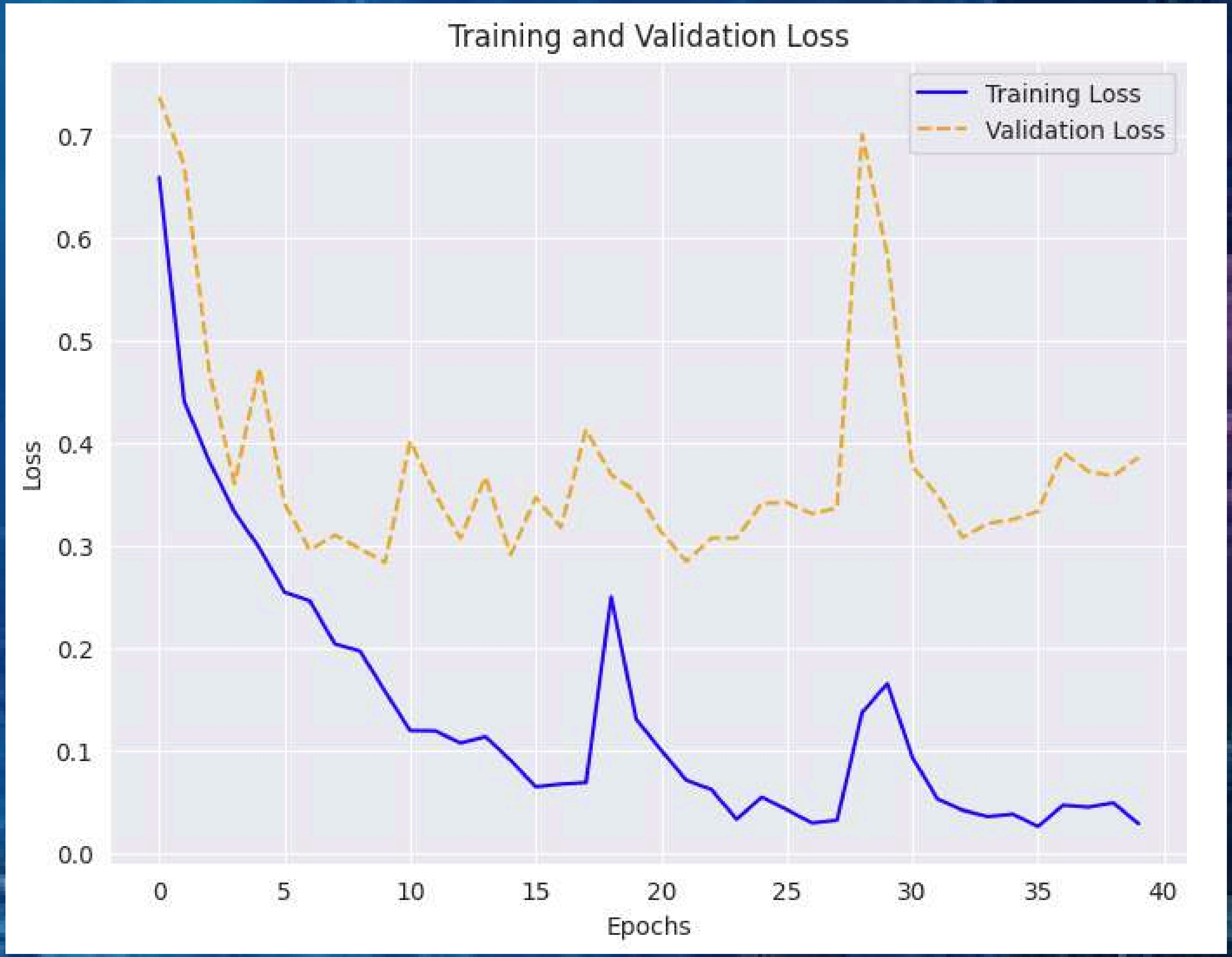
TEST SET:
  Turtle_Tortoise: 286 images
  Dolphin: 118 images
  Jelly Fish: 127 images

VAL SET:
  Turtle_Tortoise: 285 images
  Dolphin: 117 images
  Jelly Fish: 127 images
```

MODEL BUILDING

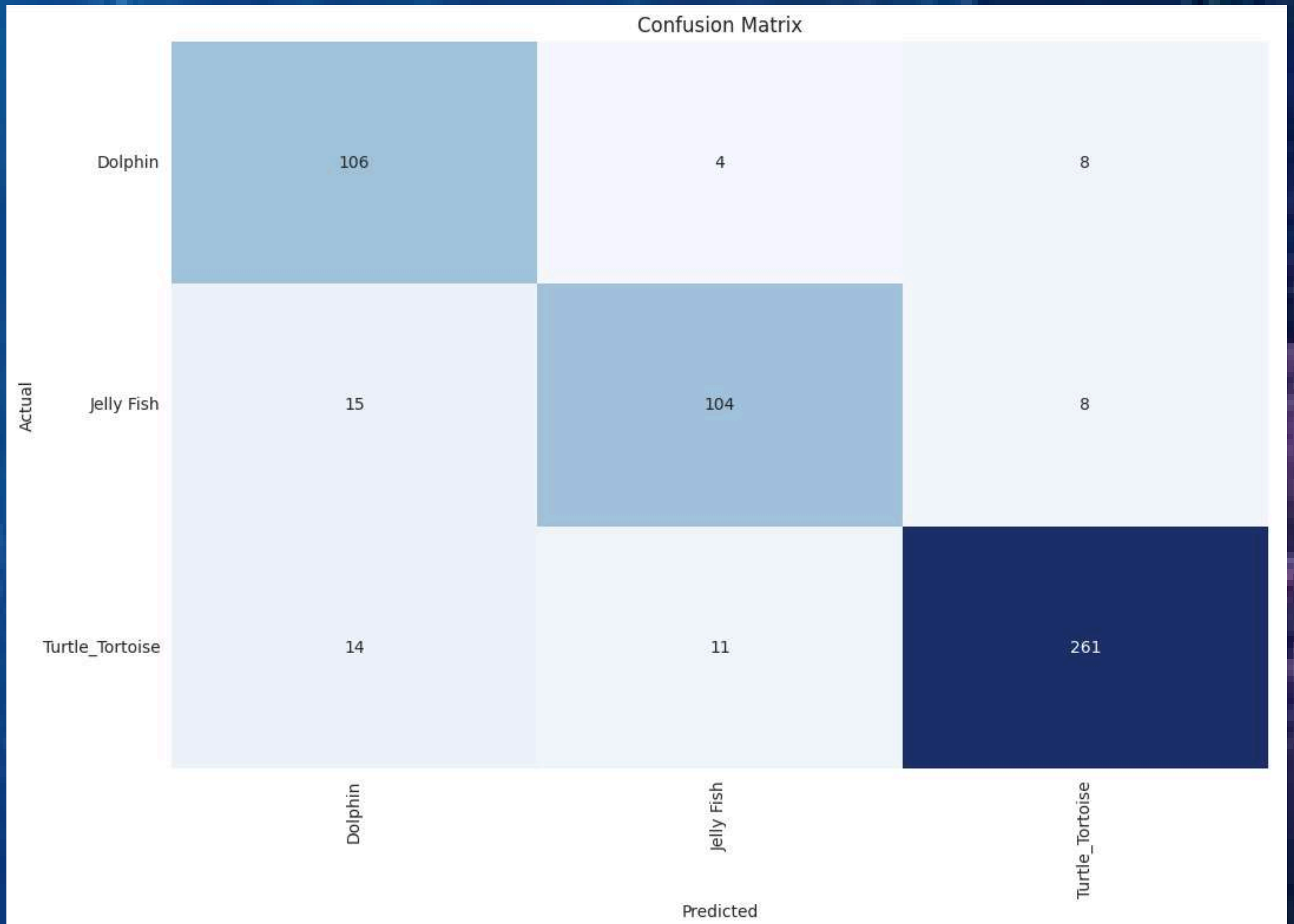
- Overview:
  - Type: Convolutional Neural Network (CNN) for multi-class image classification (3 classes).
  - Architecture:
    - Series of Conv2D layers with increasing filter sizes (32, 64, 128).
    - BatchNormalization and MaxPooling2D for feature extraction and dimensionality reduction.
    - Final dense layer with 3 units and softmax activation for multi-class classification.

- Model Compilation:
  - Optimizer: Adam
  - Loss Function: Categorical cross-entropy
  - Evaluation Metric: Accuracy



MODEL BUILDING

- Training Strategy:
    - Distributed Training: MirroredStrategy used for efficient training across multiple GPUs.
    - Optimal Performance: Achieved 99.45% accuracy on training data with 0.0325 training loss.
    - Validation: 92.06% validation accuracy and 0.3072 validation loss.
- Model Performance:
    - Strong performance in classifying Turtle/Tortoise.
    - Opportunities for improvement in classifying Dolphin and Jellyfish.



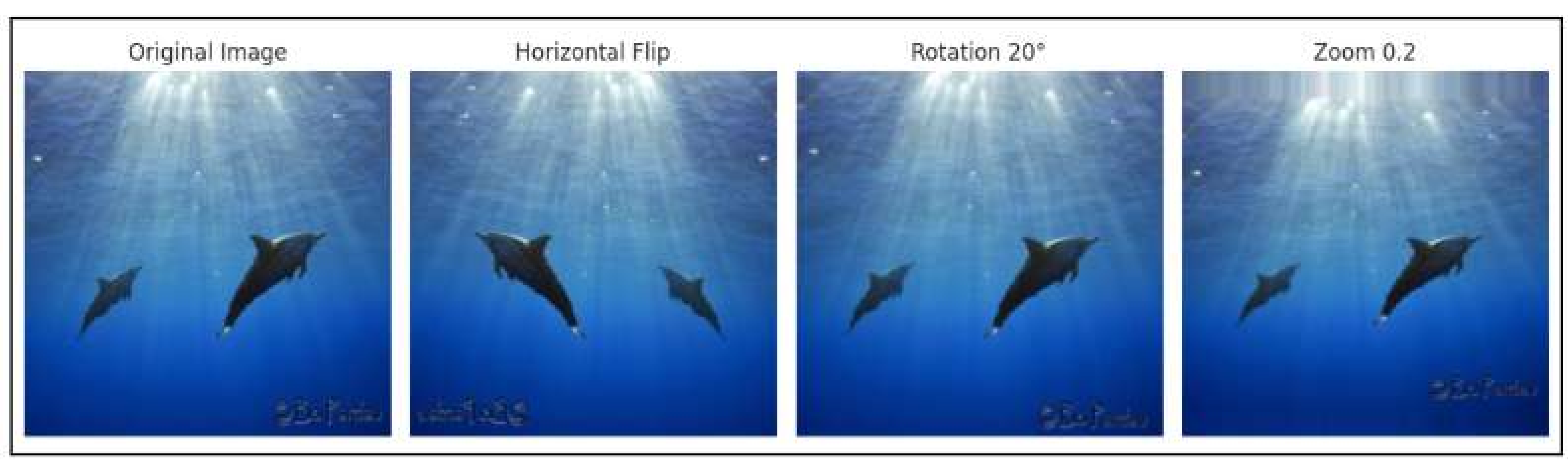
Classification Report:

	precision	recall	f1-score	support
Dolphin	0.7852	0.8983	0.8379	118
Jelly Fish	0.8739	0.8189	0.8455	127
Turtle_Tortoise	0.9422	0.9126	0.9272	286
accuracy			0.8870	531

DATA AUGMENTATION

Expanding the Dataset: Data Augmentation for Enhanced Performance

- Techniques:
  - Horizontal flipping, 20° rotation, and 0.2 zoom range applied to the training set.
  - Rescaling for validation and test sets to normalize pixel values.
  - Images resized to 224x224 for consistency and batch processing used for efficient handling.
- Training & Validation Results:
  - Epoch 34: Optimal validation performance with 93.57% validation accuracy and validation loss of 0.2122.

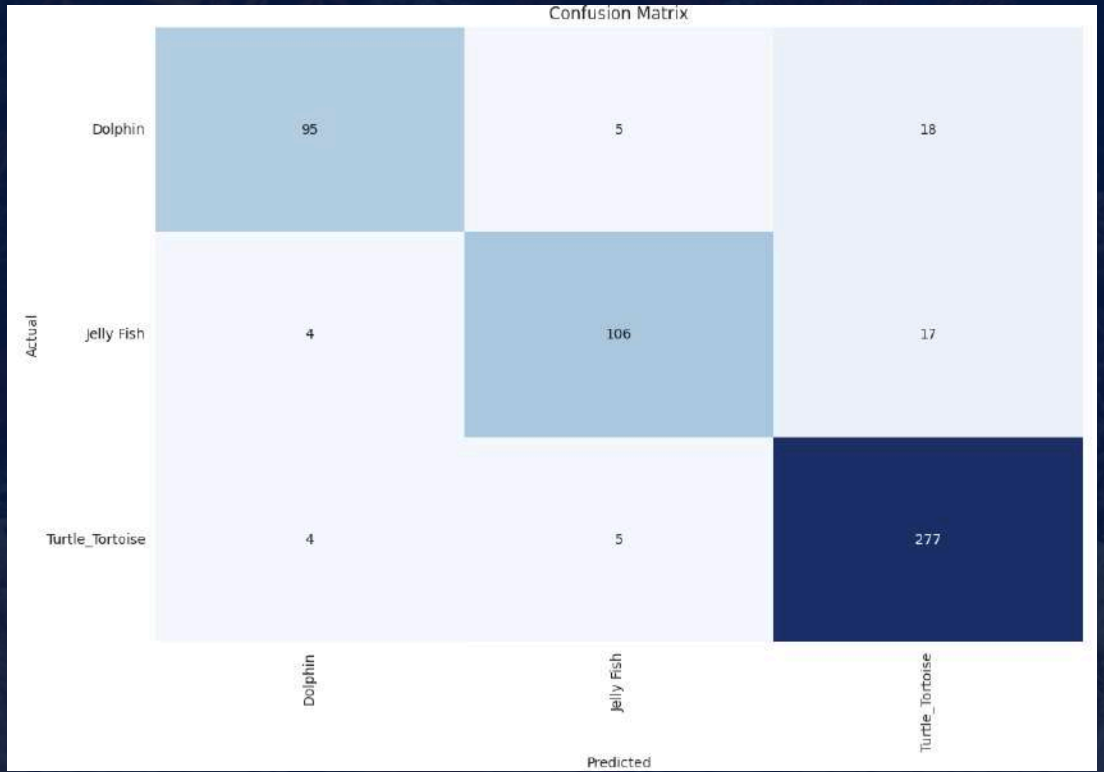


DATA AUGMENTATION

Expanding the Dataset: Data Augmentation for Enhanced Performance

- Impact of Augmentation:
    - Achieved 90.02% overall test accuracy.

Augmented model outperforms the base model, improving generalization and addressing class imbalance and overfitting.



Classification Report:

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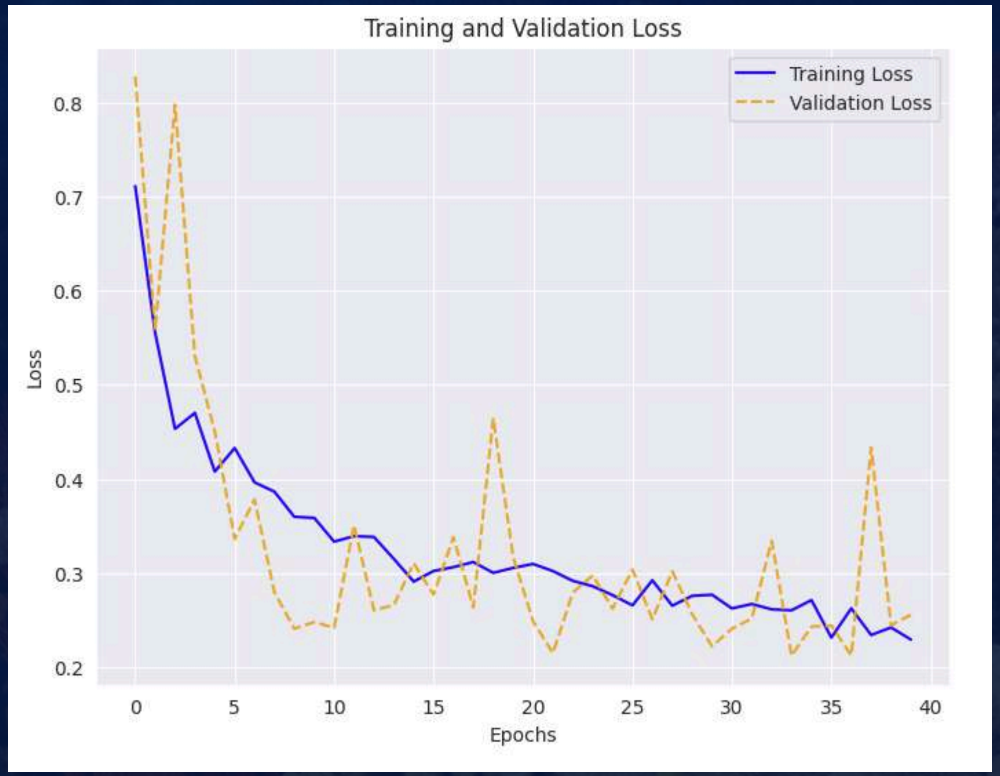
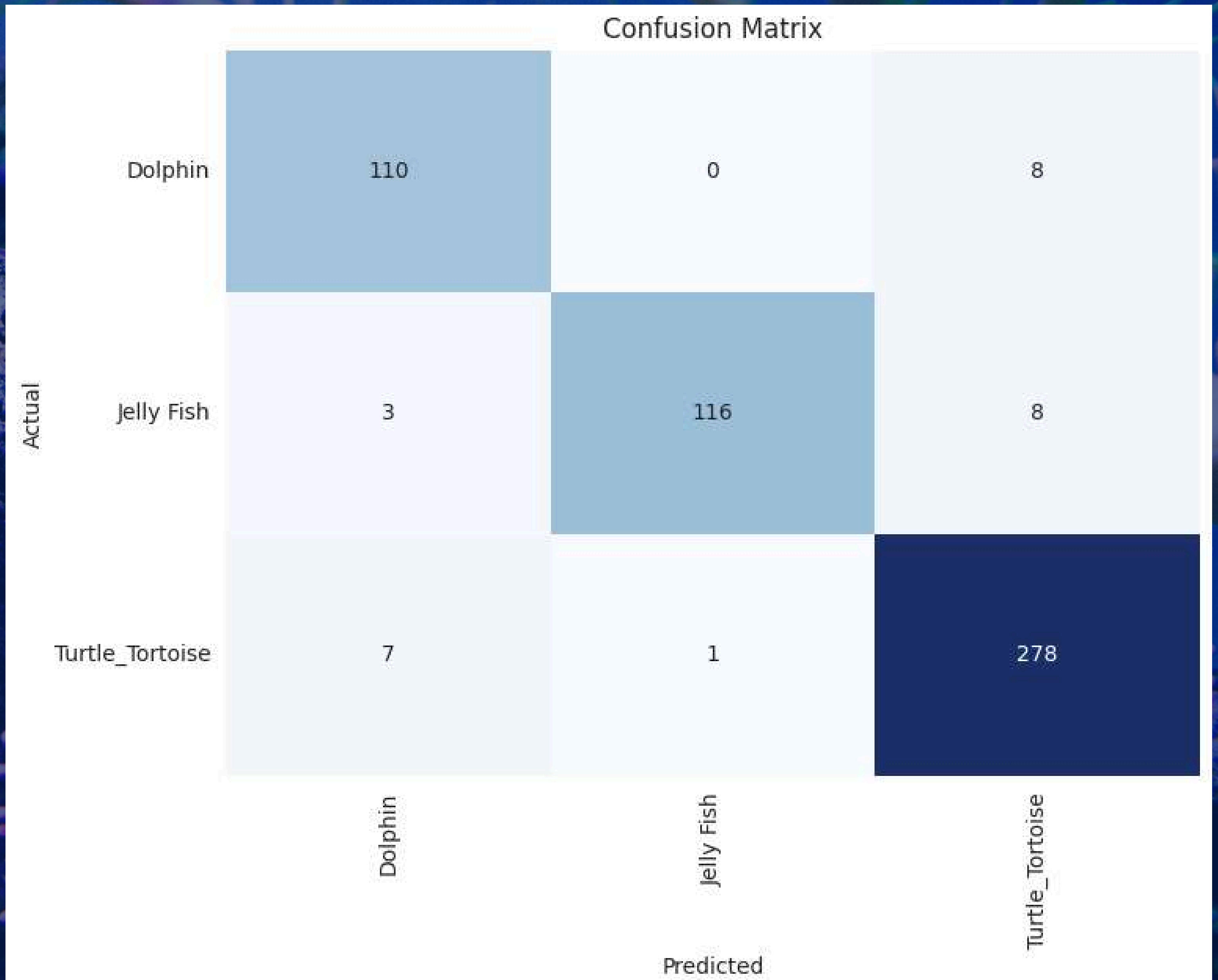
	precision	recall	f1-score	support
Dolphin	0.9223	0.8051	0.8597	118
Jelly Fish	0.9138	0.8346	0.8724	127
Turtle_Tortoise	0.8878	0.9685	0.9264	286
accuracy			0.9002	531



Transfer Learning

Leveraging Pre-trained Knowledge: Transfer Learning for Efficiency

- Transfer learning involves using a model pre-trained on a large dataset like EfficientNetB7 and ResNet50 for adapting a new, but related, task.
- This approach significantly reduced training time with the use of train decorator function and improved performance of the model.
- Achieved a 94.92% accuracy using transfer learning.



Classification Report:
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	precision	recall	f1-score	support
Dolphin	0.9167	0.9322	0.9244	118
Jelly Fish	0.9915	0.9134	0.9508	127
Turtle_Tortoise	0.9456	0.9720	0.9586	286
accuracy			0.9492	531

MODEL PERFORMANCE METRICS

Technique	Test Accuracy	Validation Loss
Initial Model	88.70%	0.3072
Data Augmentation	90.02%	0.2122
Transfer Learning	94.92%	0.183

CHALLENGES

- Initial Vision:
  - Designed a CNN for face recognition, aiming to classify age, gender, and emotion for a mood-based recommendation system.
  - Faced challenges with large datasets, resulting in computation limitations (e.g., Colab's runtime and GPU interruptions).

- Shift to Textual Data:
  - Transitioned to MBTI personality classification using LSTM and GRU models.
  - Encountered issues like overfitting and poor model performance, leading to unsatisfactory results.

- Final Iteration - Sea Animal Classification:
  - Focused on a manageable dataset (sea animals) and adopted transfer learning.
  - Addressed issues of overfitting and class imbalance with data augmentation, dropout regularization, and early stopping.

THANK YOU
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